

MUHAMMAD KHURRAM

Backend & AI/ML Developer — Python
github.com/Khurram315048

SUMMARY

Computer Science student with strong backend development skills in Python (OOP), PHP, and C++, plus hands-on AI/ML experience across Computer Vision, Generative AI, and Geospatial AI. Experienced in building secure authentication systems, LMS platforms, REST APIs, and end-to-end AI applications with FastAPI and Streamlit. Proficient in SQLAlchemy for ORM-based database interactions. Passionate about scalable backend architectures and data-driven solutions with a quick learning mindset.

EDUCATION

University of Southern Punjab — Multan, Punjab, Pakistan

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures, Algorithms, Database Systems, Object-Oriented Programming, Web Development, Parallel & Distributed Computing

TECHNICAL SKILLS

Languages: Python (OOP), PHP, C++ (OOP), SQL, JavaScript

Frameworks: FastAPI, Flask, Streamlit

Databases / ORM: MySQL, SQLite, JSON File Storage, SQLAlchemy

AI/ML: YOLOv8 (Computer Vision), LangGraph, Ollama (Local LLMs), Google Earth Engine, Sentinel-2, Roboflow, Data Annotation & Labeling, Supervised Learning, Python APIs for AI Models

Web Tech & Tools: HTML, CSS, REST APIs, Authentication Systems, LMS Development, Folium, Git, Docker (Basic), JSON, Security Best Practices

CERTIFICATIONS

- **Crash Course on Python** — Google (via Coursera) | June 8, 2026
- **Google IT Automation with Python** — Google (via Coursera) | July 12, 2026
- **Google AI Professional Certificate** — Google (via Coursera) | June 9, 2026

PROFESSIONAL EXPERIENCE

Machine Learning Intern — Punjab Information Technology Board (PITB)

August 2026 – October 2026

- Built three end-to-end AI projects spanning Geospatial AI, Computer Vision, and Generative AI under the guidance of an AI/ML Engineer mentor
- Converted experimental Jupyter Notebook code into deployable applications using FastAPI and Streamlit
- Packaged AI applications into structured repositories with managed dependencies and technical documentation

AI/ML Engineer Intern — Infinity Wave

June 2026 – August 2026

- Performed data annotation and labeling to prepare datasets for machine learning
- Trained a machine learning model that learns from the labeled data (supervised learning)

Python Developer — Robusto Solutions Pakistan

- Architected and implemented a secure Python-based authentication system with JSON file storage, featuring bcrypt password hashing and brute-force protection through login attempt tracking
- Reduced potential security vulnerabilities by 95% through implementation of industry-standard password hashing algorithms and secure session management
- Integrated multiple AI models via Python APIs, developing custom scripts for natural language processing and data analysis tasks
- Designed and optimized database schemas using SQLAlchemy ORM for efficient data retrieval, improving query performance by implementing proper indexing strategies
- Developed RESTful API endpoints for seamless frontend-backend communication with proper error handling and validation

PROJECTS

Learning Management System (LMS) — FastAPI

Python, FastAPI, MySQL, SQLAlchemy, Pydantic, REST APIs

- Developing a full-featured LMS platform with student enrollment, course management, and assignment submission modules
- Refactored the backend from a Flask and raw MySQL hybrid into a pure FastAPI architecture with SQLAlchemy ORM and Pydantic schema validation
- Built RESTful APIs for student and instructor dashboards with role-based access control (RBAC)

GeoGuardian — Geospatial AI Monitoring Dashboard

Python, FastAPI, Google Earth Engine, Sentinel-2, Folium, Dynamic World

- Built a web-based monitoring dashboard to analyze urban expansion and agricultural land loss using satellite imagery
- Moved the solution from an experimental Jupyter Notebook to a deployable FastAPI backend application
- Generates visual statistical insights on land-use change for regional analysis (e.g., Multan district)

Traffic Safety Helmet Detection System

Python, YOLOv8, Roboflow, Streamlit

- Trained and evaluated YOLOv8 object detection models to identify helmets for traffic safety applications
- Wrote Python scripts to download and manage Roboflow datasets; built an interactive Streamlit interface to demonstrate real-time model inference

AI-Powered Viral Content Generator

Python, LangGraph, Ollama (gemma3:4b)

- Designed an agent graph with LangGraph powered by a local gemma3:4b LLM running on Ollama
- Automatically generates trending content ideas, structured scripts, and targeted hashtags to streamline content creation

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Native)

Interests: Backend Systems Architecture, Distributed Systems, Cloud Computing, AI/ML, LMS & EdTech Platforms